

## Northwind Analytics - Architecture overview

### Four layers

1. Ingestion extract -> embed -> index
2. Retrieval hybrid semantic + lexical search
3. Agents planning, tools, memory
4. API gateway, auth, quotas

### Storage

Metadata in DynamoDB, vectors in S3 Vectors, parents and postings in S3.

## Hybrid retrieval

The semantic leg queries S3 Vectors (topK=100).  
The lexical leg scores BM25 over S3 postings  
( $k_1=1.2$ ,  $b=0.75$ ). Both run in parallel and are  
fused with Reciprocal Rank Fusion (RRF  $k=60$ ).

Small-to-big: only child chunks are embedded; each  
child points at a parent page, hydrated with one  
GetObject per unique parent.

Reranking is a separate, opt-in cross-encoder pass.